

A counterexample to Lemma 2.5.5 of *The Theory of Several Knowing Machines*

A gap in the printed proof of the dissertation's answer to Question (3-1-2)'

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Status.

- **Kind of result:** an explicit countermodel, checked against the rendered PDF of the dissertation. It was found by one AI system and verified by another. No independent human review.
- **Scope:** it refutes the lemma as printed. It does **not** refute the theorem the lemma supports. That theorem follows from the proposed result of note 15.

Source. S. A. Alexander, *The Theory of Several Knowing Machines*, doctoral dissertation, Ohio State University, 2013 (OhioLINK ETD [osu1366362999](#)): Definition 2.5.4 and Lemma 2.5.5 (printed pp. 68–69), and Claim 8 of the proof of Theorem 2.5.11 (p. 76).

The lemma

The relation “mod_tru”. Definition 2.5.4 defines $M_1 \equiv M_2 \text{ mod}_{tru} j$ for structures with standard first-order part. Among its conditions:

- (1)–(2) certain operators are interpreted identically;
- (3) $M_p \models K_k \varphi \rightarrow \varphi[s]$ for **every** k , formula φ and $p \in \{1, 2\}$;
- (4) the same factivity for the indexed operators;
- (5) numeral-instance invariance.

The lemma. Lemma 2.5.5 takes a standard structure M with properties (3)–(5) and an i-stratifier $+$. It forms M/i by changing **only** the interpretation of K_i :

$$M/i \models K_i \varphi[s] \iff \varphi \in L_\omega, M/i \models \varphi[s], \text{ and } M \models (K_i \varphi)^+[s].$$

It asserts that $M/i \equiv M \text{ mod}_{tru} j$ whenever $j R i$. The printed proof calls conditions 1–4 “immediate”.

Counterexample

The model. Take the one-edge relation $0 R 1$, with $j = 0$ and $i = 1$. Let M have standard arithmetic and be defined recursively by:

- $M \models K_k \varphi[s] \iff M \models \varphi[s]$ for every unstratified K_k ;
- every indexed K_k^α false everywhere.

M satisfies the lemma's hypotheses.

- M is a structure: independence from non-free variables, alphabetic invariance and weak substitution all follow by induction on formulas.
- M is numeral-instance invariant.
- Every operator is factive: the unstratified ones by definition, the indexed ones vacuously.

The failure. Let τ be $0 = 0$.

- **In M:** $M \models K_i \tau$, and therefore $M \models K_j K_i \tau$.
- **In M/i, the changed operator:** $(K_i \tau)^+$ is $K_i^\alpha \tau$ for some α , which is false in M. So $M/i \not\models K_i \tau$.
- **In M/i, the frozen operator:** the interpretation of K_j is unchanged, so its value on the argument $K_i \tau$ is still “true”: $M/i \models K_j K_i \tau$.

Hence M/i violates $K_j K_i \tau \rightarrow K_i \tau$. Condition (3) fails for j itself, contradicting the lemma. $\square$

Where the argument breaks, and consequences

The cause. Factivity of the *changed* operator follows from its new truth conjunct. Factivity of a *frozen* operator does not: some of its previously true assertions mention K_i , and K_i ’s truth value has changed. Adding a truth filter to K_j would restore factivity, but it would change K_j ’s interpretation, which conditions (1)–(2) require to stay fixed.

Effect on the dissertation. Theorem 2.5.11 applies the lemma in Claim 8 (the stratified subordinate reflection schema). So the printed proof of the dissertation’s answer to (3-1-2)’ has a gap at that step.

- The theorem’s conclusion is not refuted.
- A preservation argument tailored to the constructed theories could repair it.
- The proposed proof of the stronger Question 3-1-2 in note 15 avoids this operation entirely. It rebuilds *every* truth filter, and then *proves* that the relevant subordinate interpretation is unchanged by induction. If that proof is correct, it implies Theorem 2.5.11.

A related caution. Rebuilding *all* truth filters is not a universal fix either. It can destroy *mechanicalness* instead of factivity. Suppose a superior knows a code for a lower machine’s column on $\sigma \wedge “x \in W_x”$, where σ is true, and the lower machine is replaced by one that does not prove σ . Then the superior’s rebuilt column on the corresponding code-certificate instances is the nonhalting set. This is why note 15’s induction quantifies over *all* external interpretations, and why its method does not directly give Question 3-1-3.